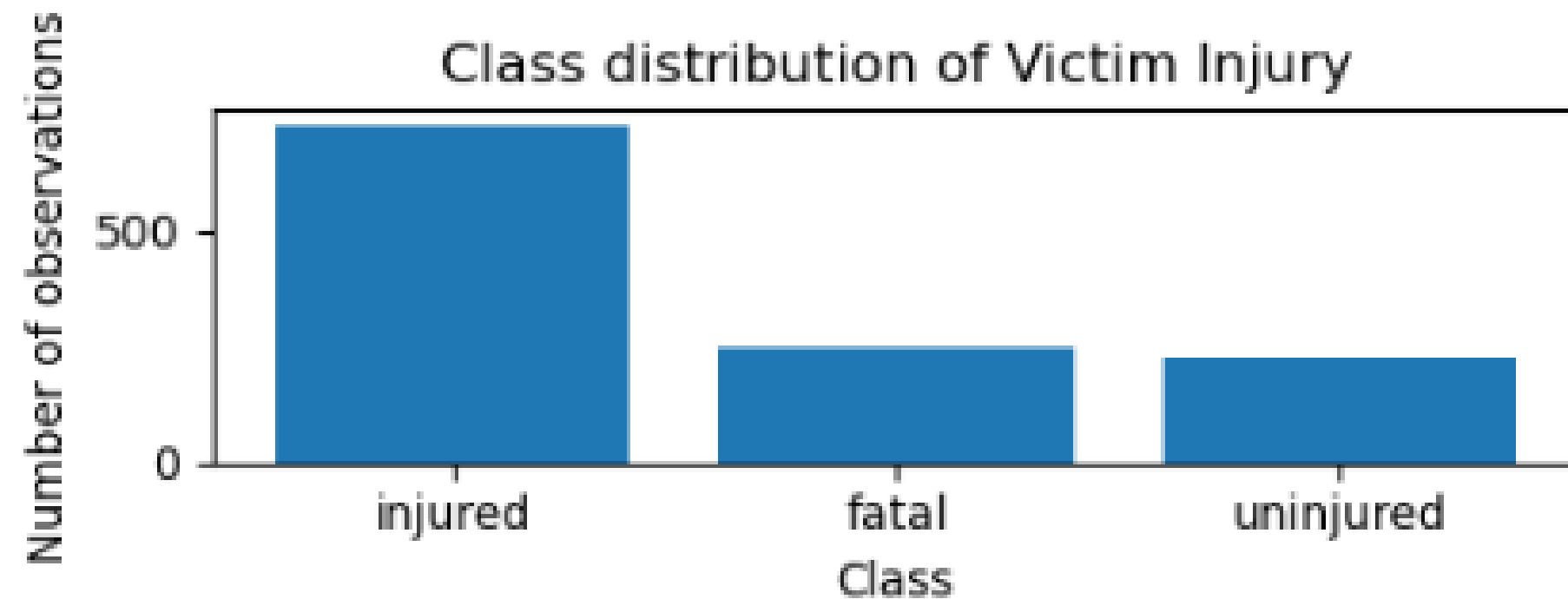


# Shark Dataset

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Taronga Conservation Society Australia  
Project #1

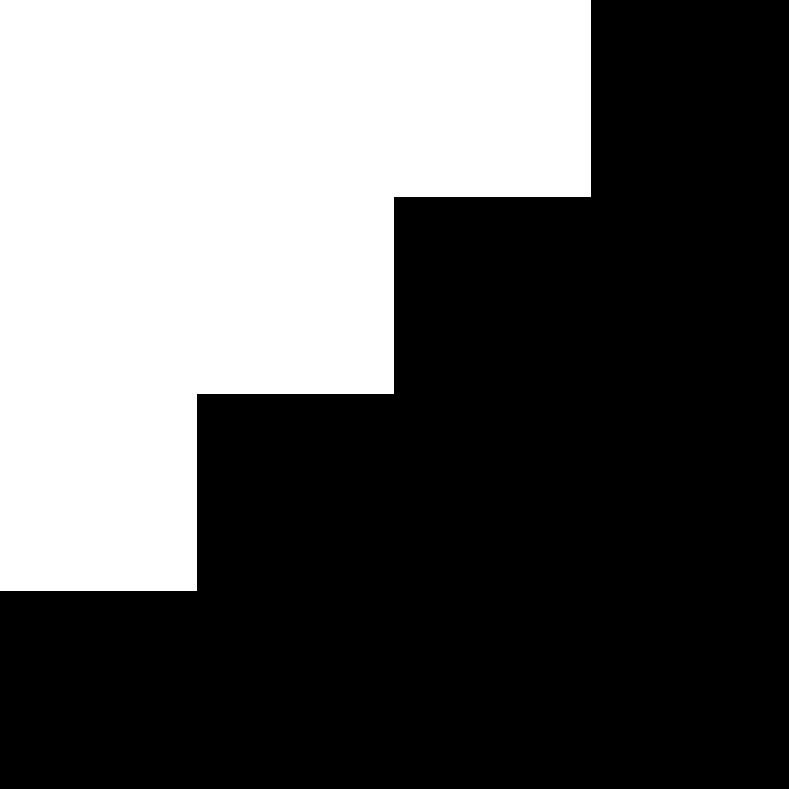
```
Victim.injury  
injured      723  
fatal        250  
uninjured    223  
Name: count, dtype: int64
```



# Problem Statement

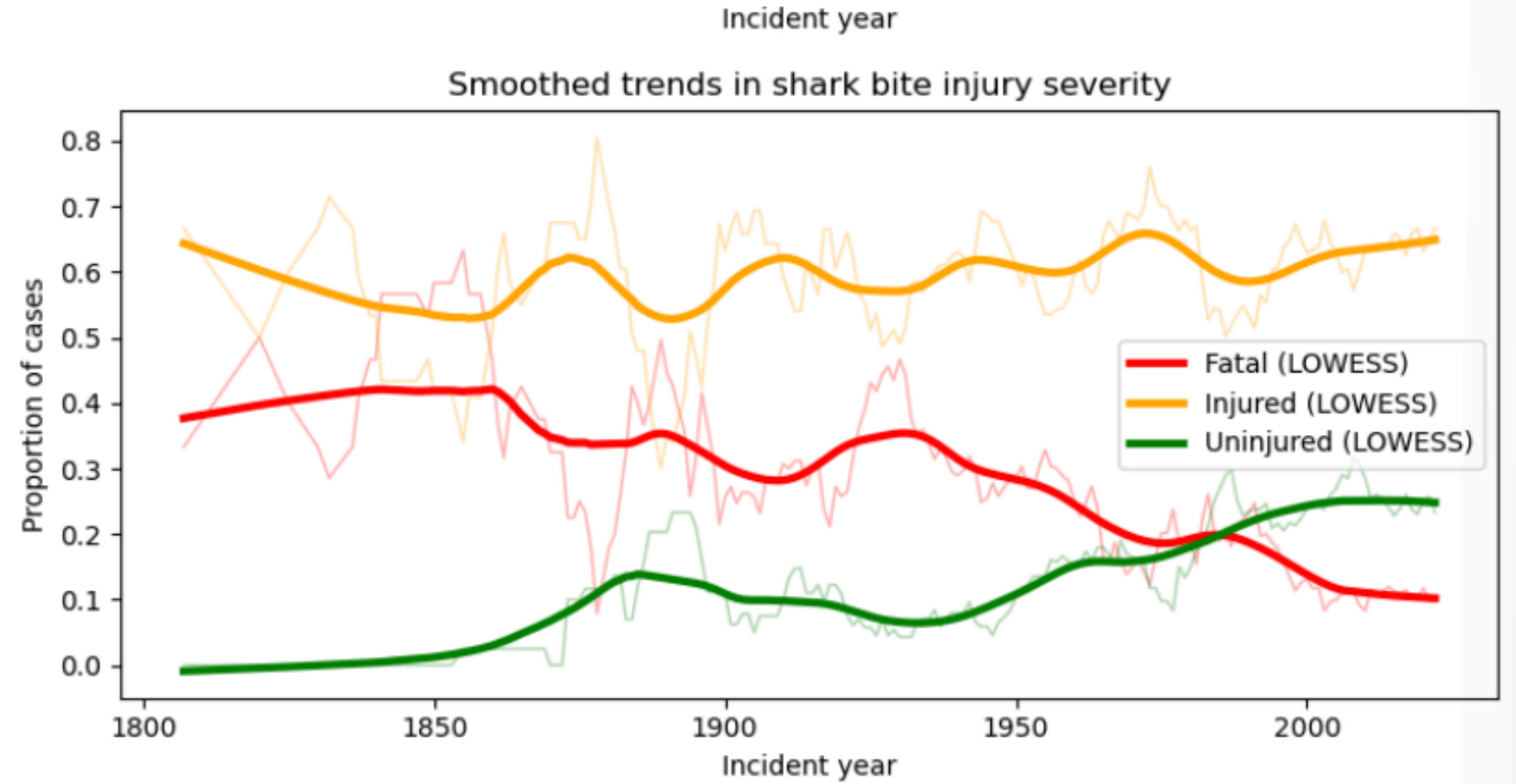
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This project aims to build a supervised machine learning model that predicts whether a shark interaction results in a fatal, injured or uninjured, using historical incident data.



# **Data Analysis**

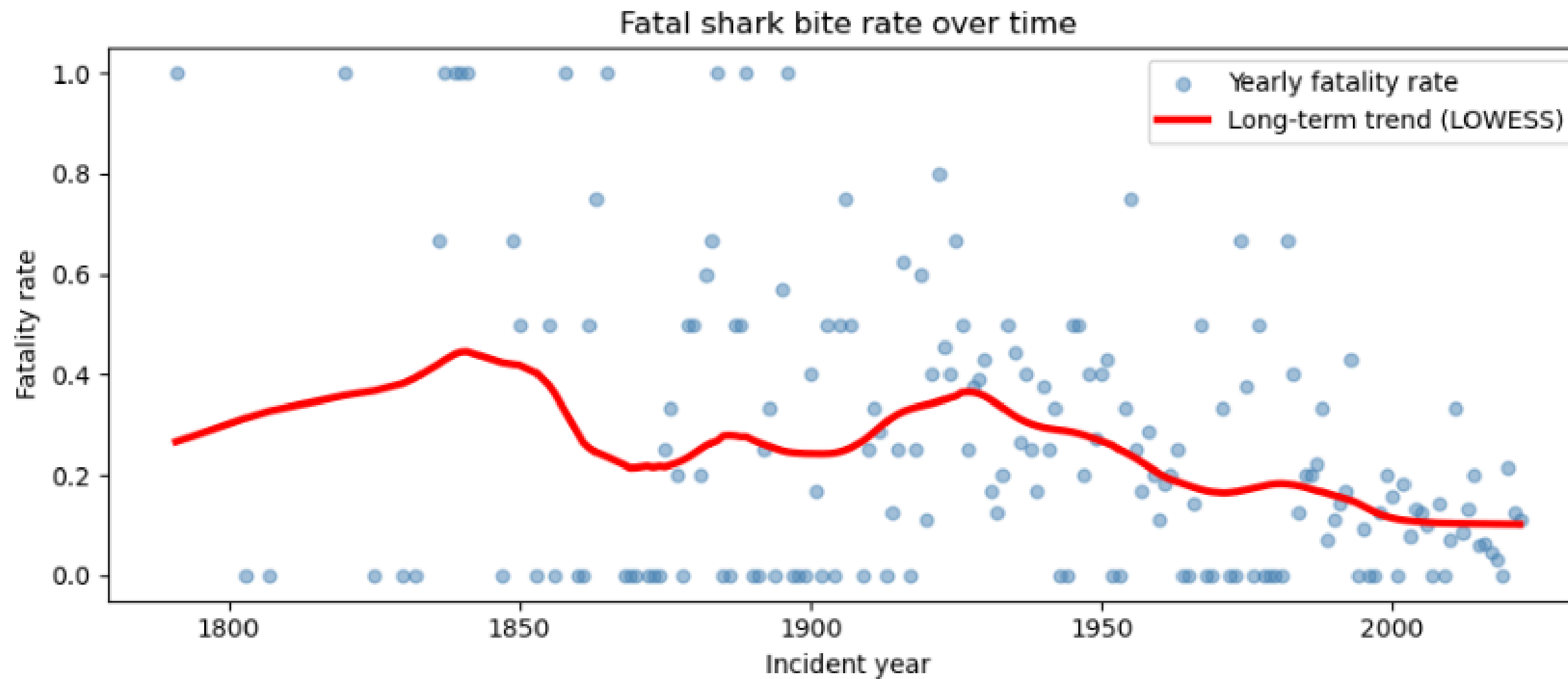
# Shark bites severity across the years



\*A smoothing parameter (frac = 0.2) was selected to emphasise broad temporal patterns while reducing short-term noise. Thin lines represent **rolling yearly proportions**, while **thick lines** represent **LOWESS-smoothed trends**.

Kruskal-Wallis H: 92.91425453226698  
p-value: 6.666931350958112e-21

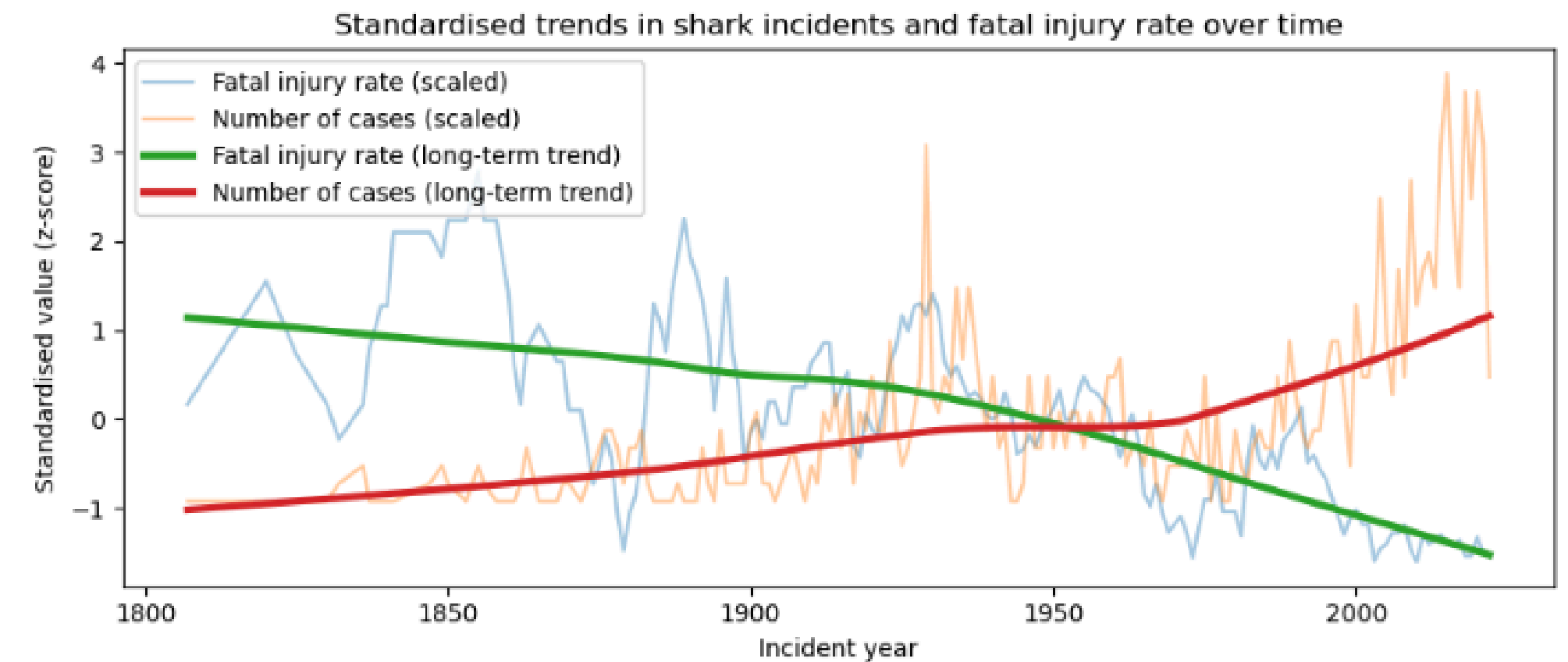
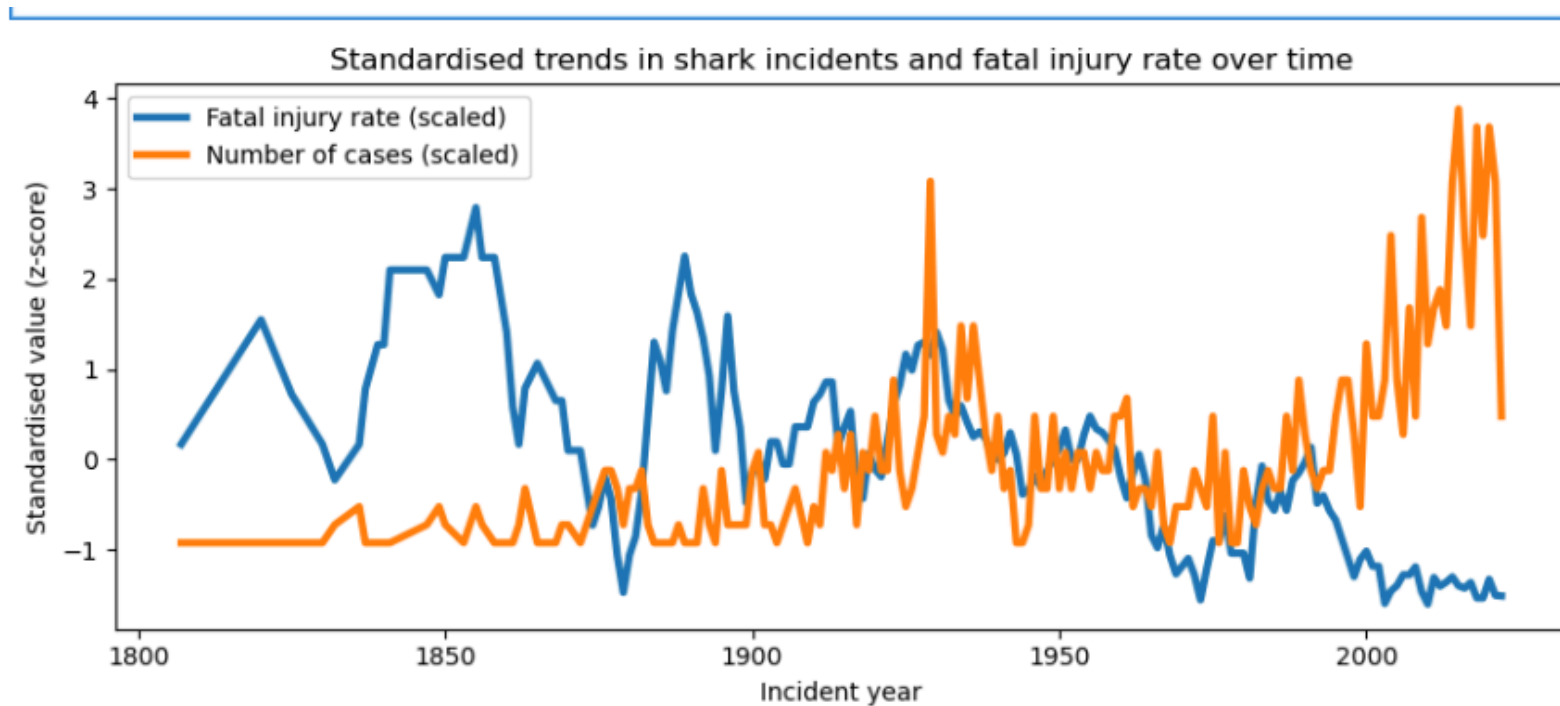
Kruskal-Wallis test revealed a statistically significant difference in incident years across injury severity categories ( $H = 92.91$ ,  $p < 0.001$ ). This indicates that the temporal distribution of shark bite outcomes differs substantially between fatal, injured, and uninjured cases.



The **declining** fatality rate observed in the LOWESS-smoothed trend is consistent with the Kruskal–Wallis test results, which showed that fatal, injured, and uninjured cases are distributed across significantly different time periods. Together, these findings **indicate** a long-term shift toward **less severe shark bite** outcomes



# Number of cases and fatal injury rate (standardised)



- Shark bite reports increase over time
- Fatality rate declines steadily
- Indicates a shift toward less severe outcomes

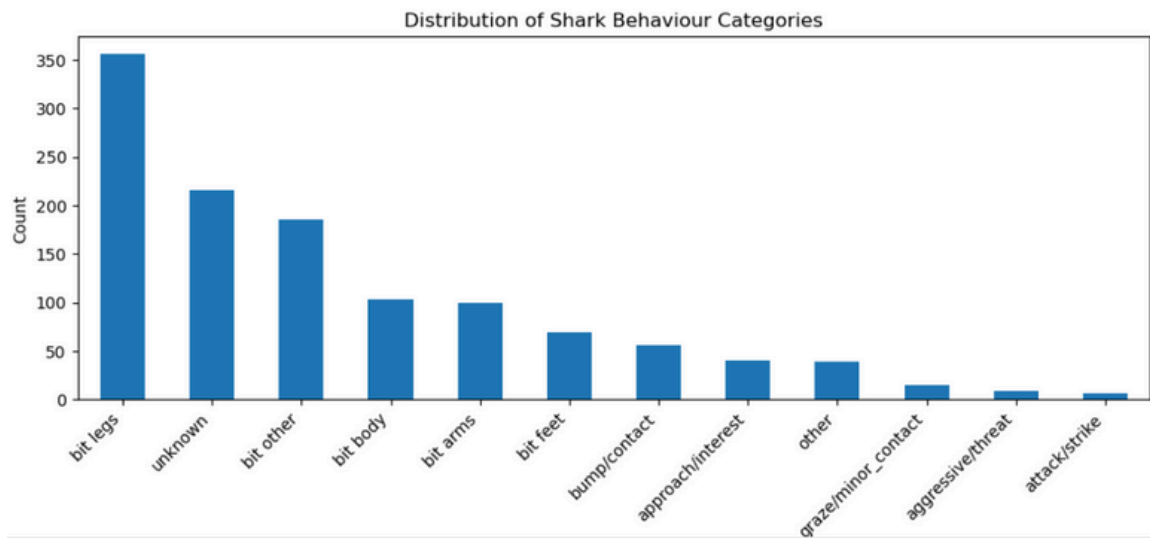
Features

Started of **60 features** (1196 entries),  
**reduced** to **23 features** (973 entries)

**Target** : Victim.injury  
(Multiclass Output: fatal vs injured vs uninjured)

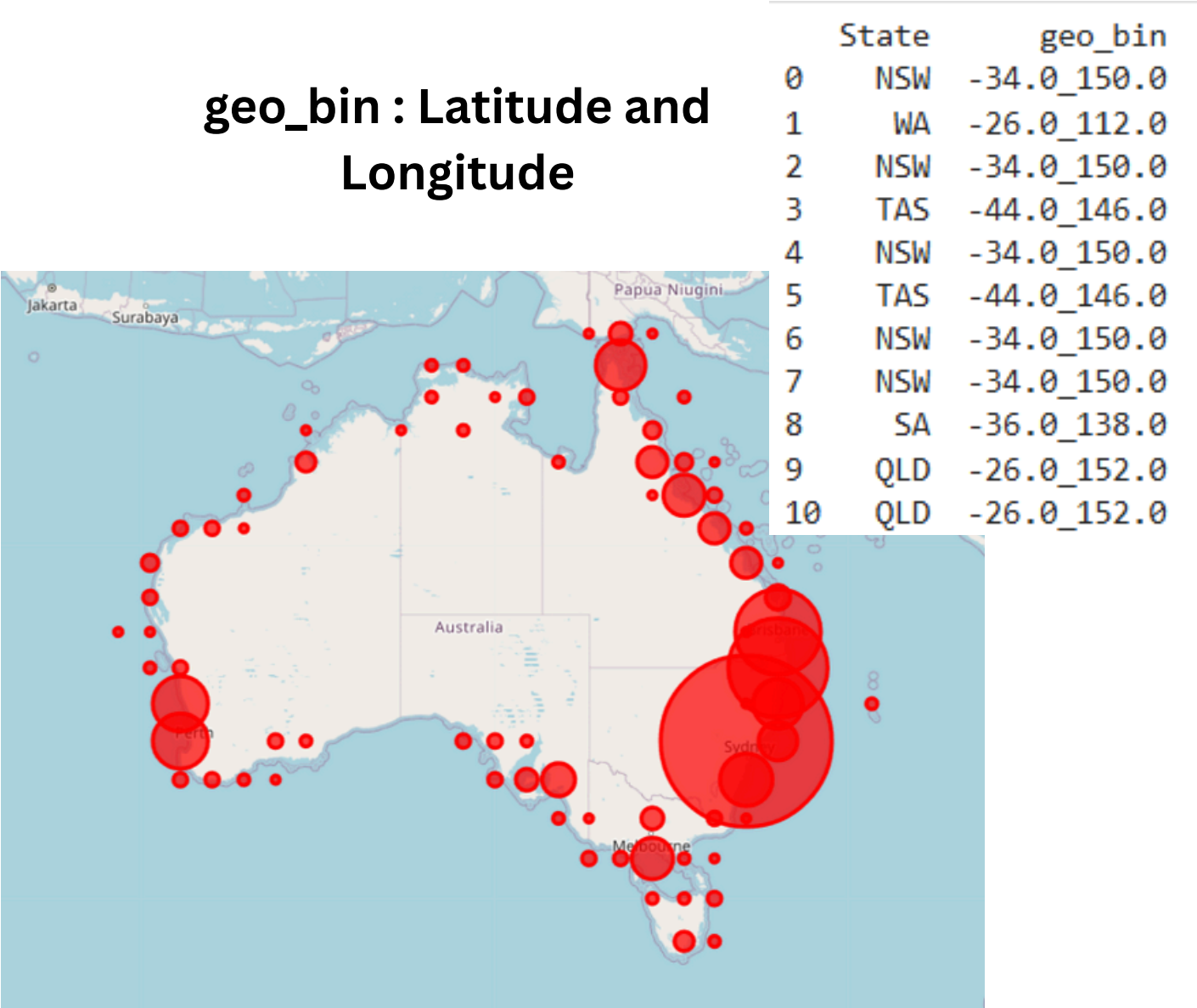
| Feature Name |                       |
|--------------|-----------------------|
| 0            | Incident.month        |
| 1            | Incident.year         |
| 2            | State                 |
| 3            | Location              |
| 4            | Site.category         |
| 5            | Shark.common.name     |
| 6            | Provoked/unprovoked   |
| 7            | Victim.activity       |
| 8            | Victim.aware.of.shark |
| 9            | Injury.location       |
| 10           | Victim.gender         |
| 11           | Victim.age            |
| 12           | geo_bin               |
| 13           | bitvsother            |

Bitvsother : Classifying shark interactions with victims



Engineered Features

geo\_bin : Latitude and Longitude



# Model1

→ **Logistic Regression (Base Model)**

AUC PR == 0.845  
(macro)

Feature Importance

|                                               | fatal     | injured   | uninjured | overall_importance |
|-----------------------------------------------|-----------|-----------|-----------|--------------------|
| cat_Injury.location_other: uninjured          | -1.811878 | -2.542095 | 4.353973  | 5.357446           |
| cat_Injury.location_other: body not recovered | 1.849522  | -1.555979 | -0.293543 | 2.434742           |
| cat_Injury.location_Unknown                   | -0.516160 | -1.287612 | 1.803773  | 2.275514           |
| cat_Shark.common.name_wobbegong               | -1.308287 | 1.222930  | 0.085357  | 1.792891           |
| cat_Injury.location_foot                      | -1.001429 | 1.398817  | -0.397388 | 1.765635           |
| cat_Shark.common.name_tiger shark             | 0.929110  | -1.085459 | 0.156349  | 1.437328           |
| cat_Shark.behaviour_bit victim on body        | 1.050227  | -0.802769 | -0.247458 | 1.344861           |
| cat_Injury.location_arm, hand                 | -0.758299 | 1.059282  | -0.300983 | 1.337044           |
| cat_Shark.common.name_white shark             | 1.014347  | -0.767006 | -0.247341 | 1.295522           |
| cat_Location_broadhurst reef, townsville      | -0.700003 | -0.266224 | 0.966227  | 1.222487           |
| cat_Location_byron bay, tallow beach          | 0.807641  | -0.774058 | -0.033583 | 1.119186           |
| cat_Location_fraser island, sunshine coast    | 0.824063  | -0.715771 | -0.108292 | 1.096875           |
| cat_Injury.location_leg, arm                  | 0.768034  | -0.731952 | -0.036082 | 1.061570           |
| cat_Injury.location_calf                      | -0.503618 | 0.856135  | -0.352517 | 1.053976           |
| cat_Shark.common.name_bull shark              | 0.852542  | -0.461829 | -0.390713 | 1.045357           |
| cat_Location_ballina, shelly beach            | 0.766096  | -0.698961 | -0.067135 | 1.039210           |

Confusion Matrix:

```
[[ 32  13   5]
 [ 31 110   4]
 [  0   1  44]]
```

Classification Report:

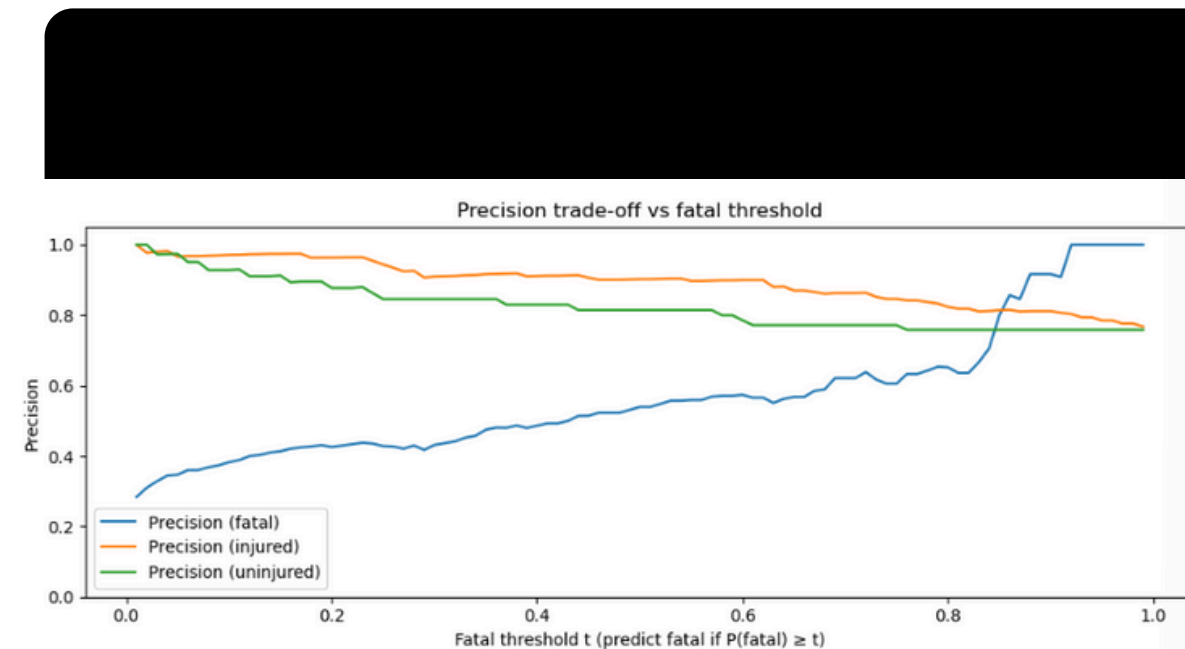
|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| fatal        | 0.51      | 0.64   | 0.57     | 50      |
| injured      | 0.89      | 0.76   | 0.82     | 145     |
| uninjured    | 0.83      | 0.98   | 0.90     | 45      |
| accuracy     |           |        | 0.78     | 240     |
| macro avg    | 0.74      | 0.79   | 0.76     | 240     |
| weighted avg | 0.80      | 0.78   | 0.78     | 240     |



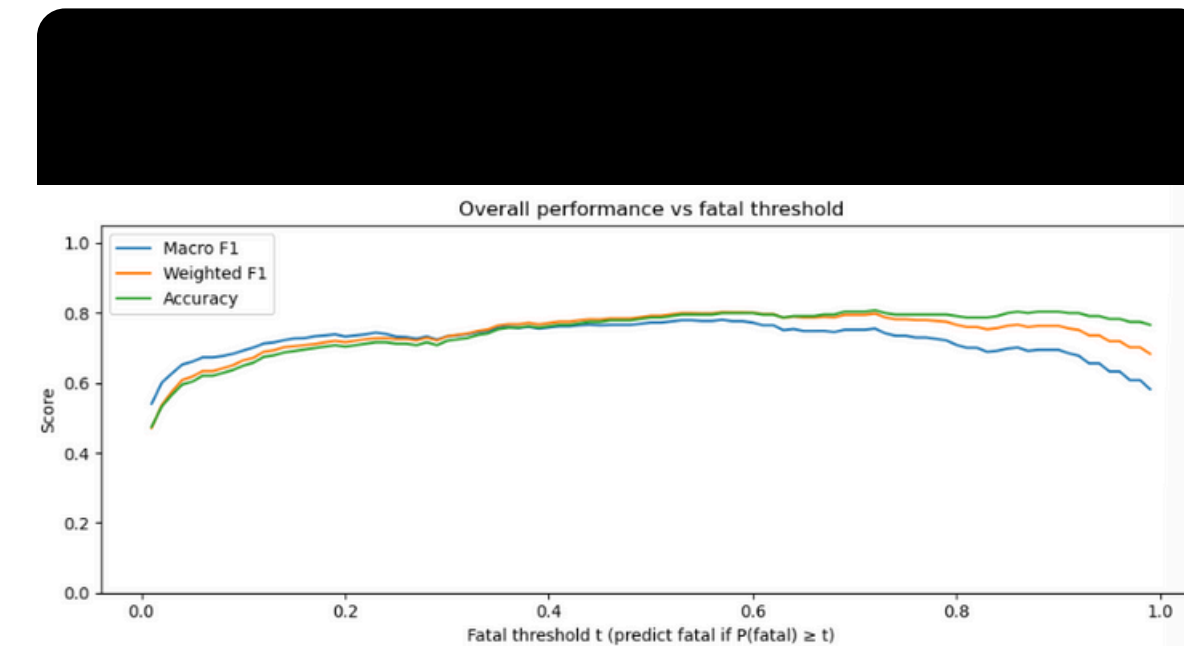
# PRECISION-RECALL vs Threshold



Recall trade offs



Precision trade offs



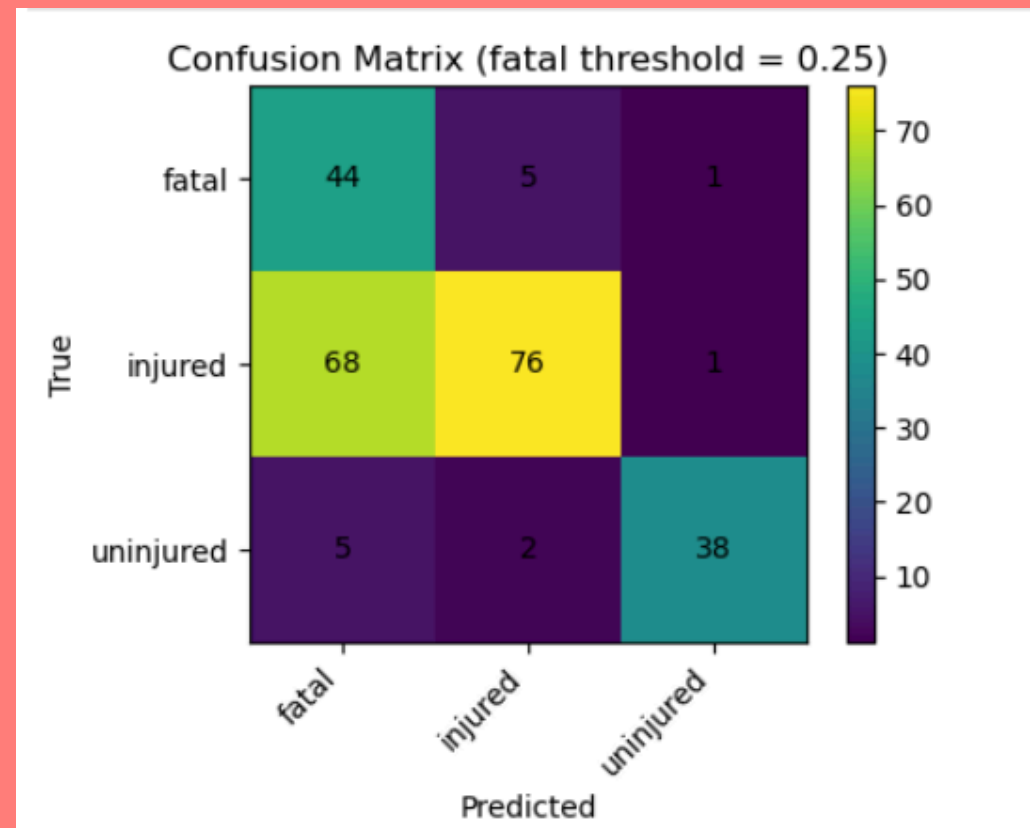
Overall trade offs

Chosen threshold : 0.25

# Model2

→ Logistic Regression (Threshold =0.25)

**F1 MACRO = 0.737**



Confusion Matrix (thr=0.25):

```
[[42  4  4]
 [53 88  4]
 [ 1  1 43]]
```

Classification Report (thr=0.25):

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| fatal        | 0.44      | 0.84   | 0.58     | 50      |
| injured      | 0.95      | 0.61   | 0.74     | 145     |
| uninjured    | 0.84      | 0.96   | 0.90     | 45      |
| accuracy     |           |        | 0.72     | 240     |
| macro avg    | 0.74      | 0.80   | 0.74     | 240     |
| weighted avg | 0.82      | 0.72   | 0.73     | 240     |

Accuracy: 0.7208333333333333

F1 macro: 0.7368905324686953

F1 weighted: 0.734610475183224

# Victim.aware.of.shark Missingness (48.6%)

| Victim.injury | fatal    |          |          | injured |  | uninjured |  |
|---------------|----------|----------|----------|---------|--|-----------|--|
| aware_missing |          |          |          |         |  |           |  |
| 0             | 0.244300 | 0.576547 | 0.179153 |         |  |           |  |
| 1             | 0.171821 | 0.634021 | 0.194158 |         |  |           |  |

assuming missing awareness = lower  
fatal risk, not missing = higher fatal risk

This indicates informative missingness  
rather than random absence.

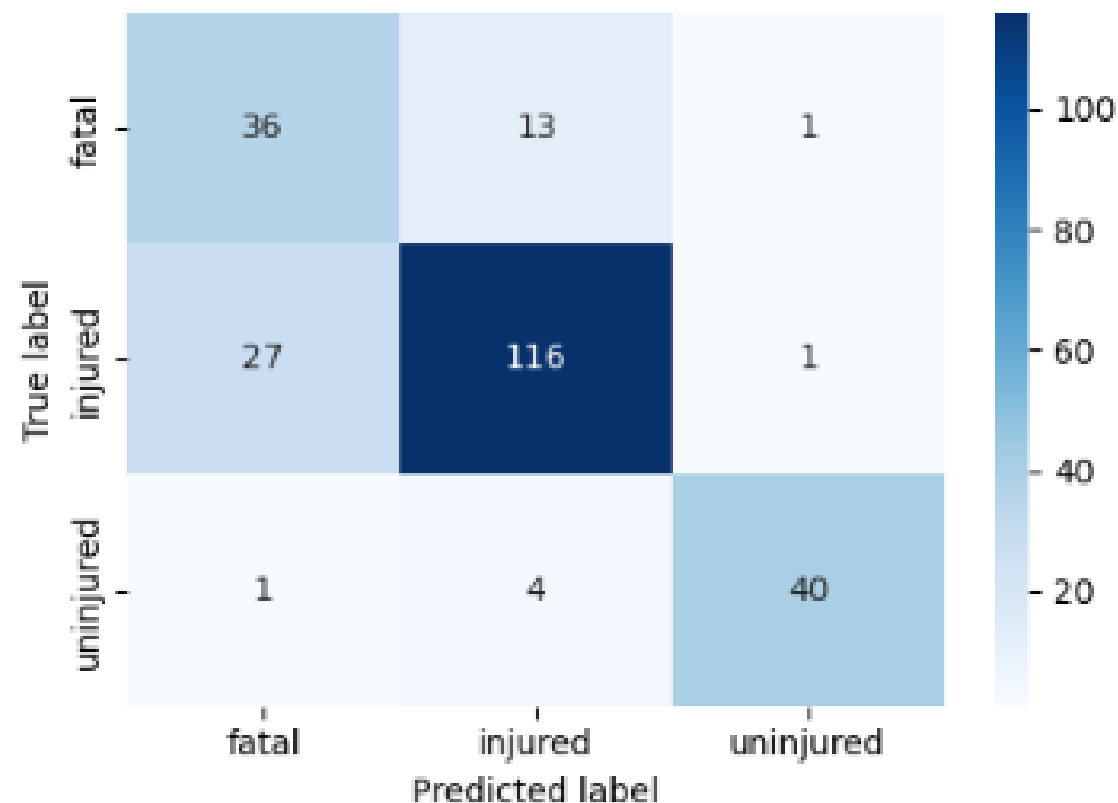
Although inclusion of Victim.aware.of.shark improved probability-based metrics (ROC-AUC and AUC-PR), it did not improve—and slightly degraded—threshold-based classification performance as measured by macro-F1 and class-specific recall. Given the primary objective of accurate injury severity classification and the desire to avoid learning dataset artefacts, this feature was excluded from the final model.

# Model2 (reduced feature)

Victim.aware.of.shark Missingness (48.6%)

| Victim.injury |  | fatal    | injured  | uninjured |
|---------------|--|----------|----------|-----------|
| aware_missing |  |          |          |           |
| 0             |  | 0.244300 | 0.576547 | 0.179153  |
| 1             |  | 0.171821 | 0.634021 | 0.194158  |

Logistic Regression (without Victim.aware.of.shark)  
Confusion Matrix



→ **Logistic Regression**  
**(Threshold =0.25)**

→ Without

Victim.aware.of.shark

AUC PR= 0.879

**F1 MACRO improved!**  
**0.796**

=== WITHOUT Victim.aware.of.shark ===

Confusion Matrix:

```
[[ 36  13   1]
 [ 27 116   1]
 [   1   4  40]]
```

Classification Report:

|              | precision | recall | f1-score |
|--------------|-----------|--------|----------|
| fatal        | 0.56      | 0.72   | 0.63     |
| injured      | 0.87      | 0.81   | 0.84     |
| uninjured    | 0.95      | 0.89   | 0.92     |
| accuracy     |           |        | 0.80     |
| macro avg    | 0.80      | 0.80   | 0.80     |
| weighted avg | 0.82      | 0.80   | 0.81     |

Model class order: ['fatal' 'injured' 'uninjure

ROC-AUC (macro, OVR): 0.9327290161256521

AUC-PR (macro): 0.8785827211145886

AUC-PR (weighted): 0.9002423208058177

# StratifiedKFold

**F1 Macro** = most important metric

**STD(0.24)** = stable across folds

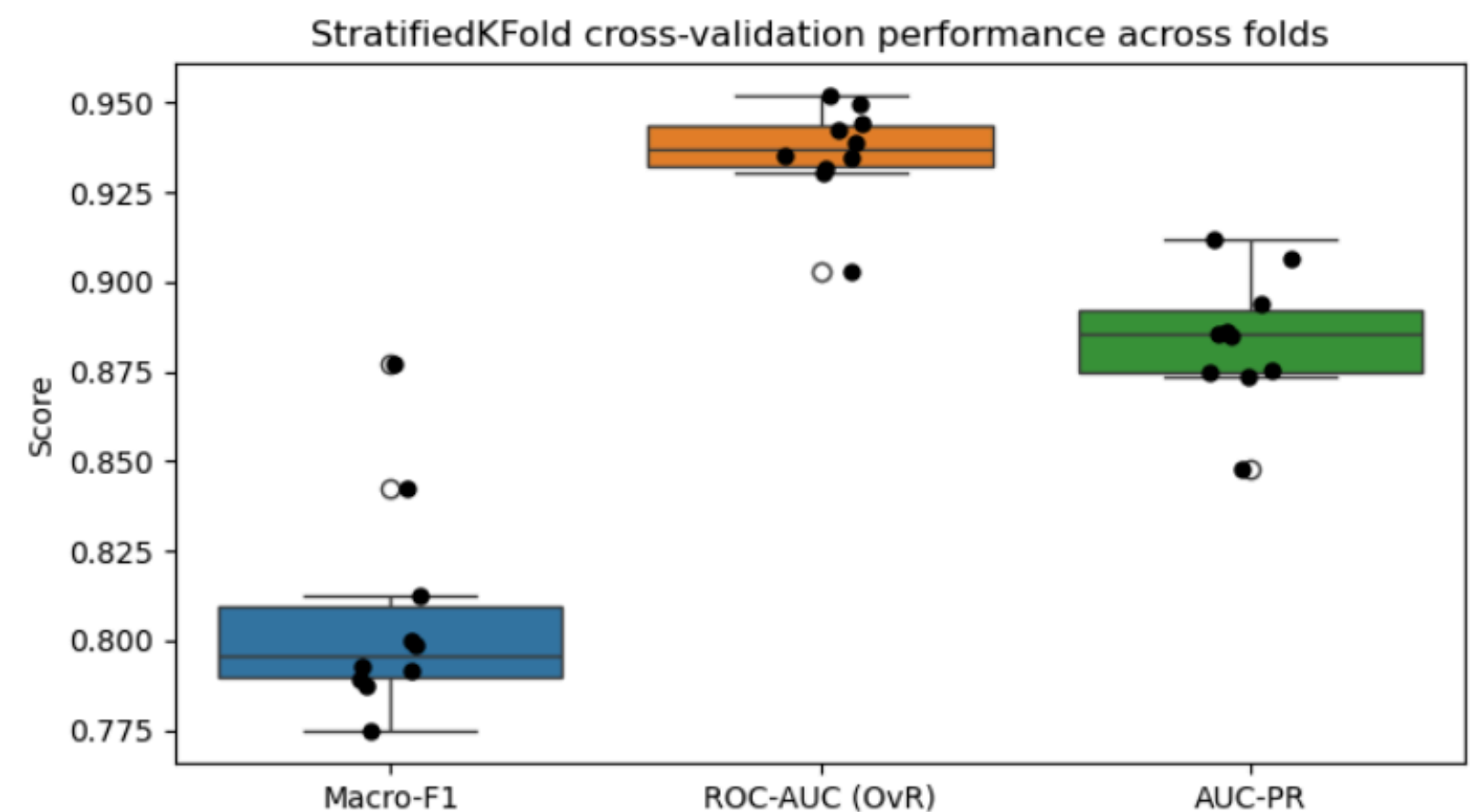
**ROC-AUC**= Strong ranking consistency

**AUC-PR** = strong performance on minority classes

```
test_f1_macro: mean=0.8132, std=0.0340
```

```
test_roc_auc_ovr_macro: mean=0.9391, std=0.0073
```

```
testauprc_macro: mean=0.8872, std=0.0176
```





# Model3

→ **XGBoost**

Include the feature  
“Victim.aware.of.shark”, helps with  
tree-like models like XGBoost.

XGB plain (with sample weight)

F1 Macro = 0.745

|           | precision | recall | f1-score | support |
|-----------|-----------|--------|----------|---------|
| fatal     | 0.67      | 0.36   | 0.47     | 50      |
| injured   | 0.81      | 0.93   | 0.87     | 145     |
| uninjured | 0.89      | 0.91   | 0.90     | 45      |

XGB With Threshold 0.07

(with sample weight)

F1 Macro = 0.745

|           | precision | recall | f1-score | support |
|-----------|-----------|--------|----------|---------|
| fatal     | 0.38      | 0.88   | 0.53     | 50      |
| injured   | 0.92      | 0.52   | 0.67     | 145     |
| uninjured | 0.95      | 0.84   | 0.89     | 45      |

# Stratified KFold

XGBoost without threshold has better overall measurements. However, recall for “Fatal” class is too low for its importance.

|                       |                    |      |      |      |    |
|-----------------------|--------------------|------|------|------|----|
| Plain (no threshold): | <code>fatal</code> | 0.71 | 0.48 | 0.57 | 50 |
| Thresholded XGBoost:  | <code>fatal</code> | 0.38 | 0.88 | 0.53 | 50 |

```
===== CV SUMMARY (MEAN over folds) =====
Mean F1 macro (plain): 0.7926561203486271
Mean F1 macro (thr=0.07): 0.7148593476458334
Mean F1 weighted (thr=0.07): 0.6983163591755324
```

| Metric             | Plain XGB (sample weight) | Thresholded XGB (sample weight + fatal threshold) |
|--------------------|---------------------------|---------------------------------------------------|
| Macro Precision    | 0.84                      | 0.767                                             |
| Macro Recall       | 0.756                     | 0.753                                             |
| Weighted Precision | 0.827                     | 0.822                                             |
| Weighted Recall    | 0.828                     | 0.674                                             |

The threshold-adjusted XGB model shows lower overall precision and weighted recall, reflecting an intentional trade-off to prioritise detection of fatal cases

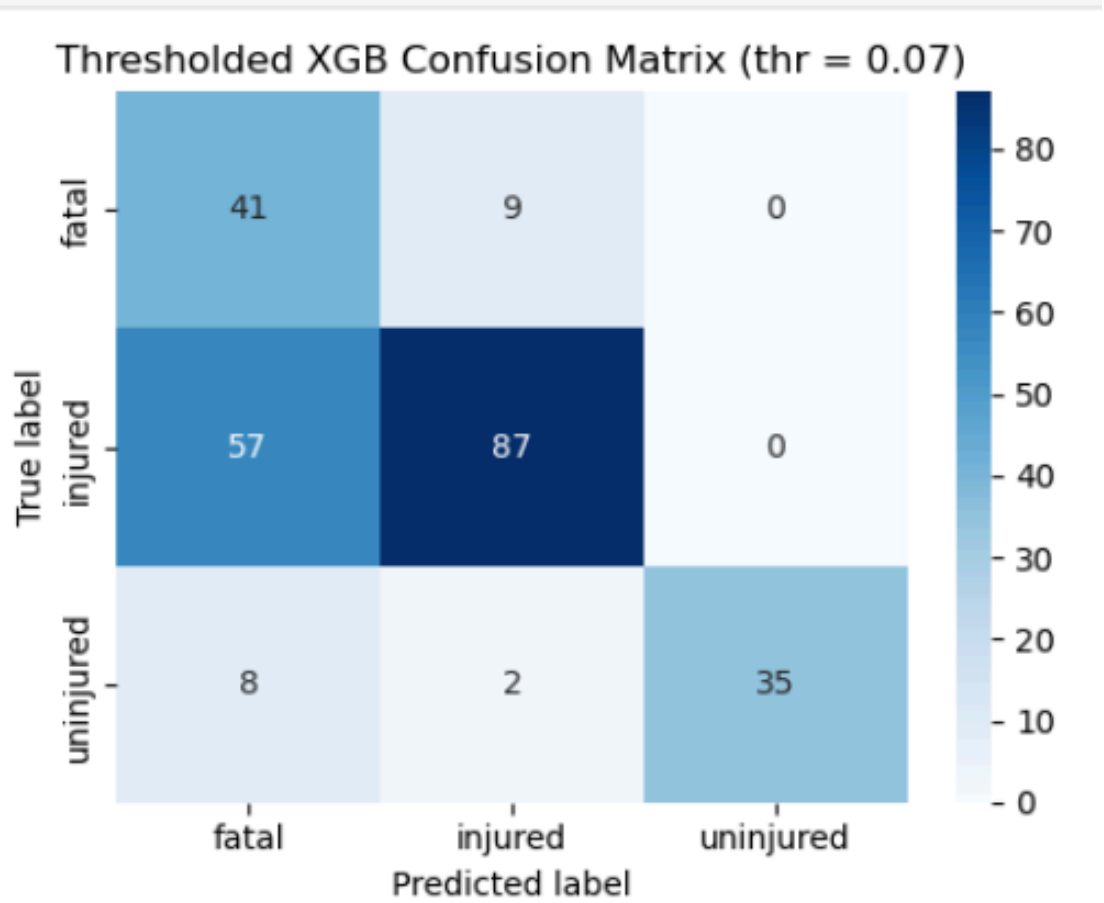
# Tuning Hyperparameter

| Item                            | Value |
|---------------------------------|-------|
| Best Macro-F1                   | 0.799 |
| Number of Estimators            | 1200  |
| Max Depth                       | 4     |
| Learning Rate                   | 0.03  |
| Min Child Weight                | 5     |
| Subsample                       | 0.8   |
| Column Sample (by tree)         | 0.9   |
| Gamma                           | 0     |
| L1 Regularisation ( $\alpha$ )  | 0.01  |
| L2 Regularisation ( $\lambda$ ) | 2     |

=== Best XGB (thresholded, thr=0.07) ===

```
[[41  9  0]
 [57 87  0]
 [ 8  2 35]]
```

|              | precision | recall | f1-score |
|--------------|-----------|--------|----------|
| fatal        | 0.39      | 0.82   | 0.53     |
| injured      | 0.89      | 0.60   | 0.72     |
| uninjured    | 1.00      | 0.78   | 0.88     |
| accuracy     |           |        | 0.68     |
| macro avg    | 0.76      | 0.73   | 0.71     |
| weighted avg | 0.80      | 0.68   | 0.71     |





# Chosen Model

## Logistic Regression with Threshold Tuning (without “Victim.aware.of.sharks”)

*Model performance was evaluated using 5-fold Stratified Cross-Validation to preserve class imbalance across folds.*

Although thresholded XGBoost substantially improves recall for fatal cases, this comes at the cost of reduced overall stability and interpretability. Therefore, the logistic regression model without Victim.aware.of.shark was selected as the final model due to its balanced macro-level performance and transparent decision structure.

Classification Report:

|           | precision | recall | f1-score | Mean ± Std: |                   |              |                 |          |
|-----------|-----------|--------|----------|-------------|-------------------|--------------|-----------------|----------|
|           |           |        |          |             | ROC_AUC_macro_ovr | AUC_PR_macro | AUC_PR_weighted | F1_macro |
| fatal     | 0.59      | 0.76   | 0.66     | mean        | 0.939594          | 0.883184     | 0.905898        | 0.822674 |
| injured   | 0.91      | 0.81   | 0.86     | std         | 0.014779          | 0.022525     | 0.019847        | 0.026039 |
| uninjured | 0.93      | 0.96   | 0.95     |             |                   |              |                 |          |